# YINGAO (ELAINE) YAO

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https://elaineyao.github.io/

## RESEARCH INTERESTS

My research interests are at the intersection of Software Engineering and Machine Learning (ML). I am particularly interested in developing software testing techniques (e.g., fuzzing) to improve the security, reliability, and correctness of ML systems.

## EDUCATION

**Cornell University** *September 2024 - Present*
Ph.D. in Computer Science.
Advisor: Dr. Saikat Dutta.

**University of British Columbia** *September 2021 - December 2023*
MASc. Electrical and Computer Engineering.
Advisor: Dr. Karthik Pattabiraman.

**University of Electronic Science and Technology of China** *September 2017 - June 2021*
B.E. Electrical and Communication Engineering
GPA: 3.96/4.0

## PUBLICATIONS

**Automated Assertion Generation and Regression Testing for Machine Learning Notebooks** *ASE 2026*
**Yingao (Elaine) Yao**, Vedant Nimje, Varun Viswanath, and Saikat Dutta.
IEEE/ACM International Conference on Automated Software Engineering (ASE).
**ACM SIGSOFT Distinguished Paper Award**

**ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic** *NDSS 2027*
Gargi Mitra, Chanyuan Liu, Pritam Dash, **Yingao Yao**, Alain Zhiyanov, Aastha Mehta, Karthik Pattabiraman.
Network and Distributed System Security (NDSS).

**SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms** *DSN 2023*
**Yingao (Elaine) Yao**, Pritam Dash, and Karthik Pattabiraman.
IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

**Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms** *CCS 2023*
**Yingao (Elaine) Yao**, Pritam Dash, and Karthik Pattabiraman.
ACM SIGSAC Conference on Computer and Communications Security (CCS).

## HONORS AND AWARDS

**ACM SIGSOFT Distinguished Paper Award** *2026*
**IEEE DSN Student Travel Grant Award** *2023*
**Faculty of Applied Science Graduate Award, University of British Columbia** *2022*
**Outstanding Graduate, Sichuan Province** *2021*
**Outstanding Winner, COMAP Interdisciplinary Contest in Modeling** *2020*
**Thanksgiving Scholarship for Modern Scientists** *2020*
**National Scholarship** *2020*
**National Second Prize in China Undergraduate Mathematical Contest in Modeling** *2018*

## WORKING EXPERIENCE

**Security Researcher** *Jan 2024 - Aug 2024*
Fortinet, Vancouver, Canada.

## TEACHING EXPERIENCE

**Teaching Assistant for CS 3780/5780: Intro to Machine Learning** *Fall 2026*
Cornell University, Ithaca, NY, USA

**Teaching Assistant for CS 5150: Software Engineering** *Spring 2025*
Cornell University, Ithaca, NY, USA

**Teaching Assistant for CPEN 321: Software Engineering** *Spring 2022, Fall 2022, Spring 2023*
University of British Columbia, Vancouver, Canada.

## ACADEMIC SERVICE

**Artifact Evaluation Program Committee** *2024*
19th USENIX Symposium on Operating Systems Design and Implementation (OSDI).
European Conference on Object-Oriented Programming (ECOOP).
18th USENIX WOOT Conference on Offensive Technologies (WOOT).
ACM Conference on Computer and Communications Security (CCS).
USENIX Annual Technical Conference (USENIX ATC).
The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

**Artifact Evaluation Program Committee** *2023*
The 34th IEEE International Symposium on Software Reliability Engineering (ISSRE).

## DEPARTMENT SERVICE

**Department of Computer Science, Picnic Czar** *Fall 2025,* Spring 2026
Cornell University, Ithaca, NY.

## MENTORING

David Jojic (BS, University of Belgrade)
Arpan Chakraborty (BS, Adamas University)
Vedant Ramesh Nimje (BS, Veermata Jijabai Technological Institute, Mumbai)
Varun Viswanath (BS, Dwarkadas J Sanghvi College of Engineering)
Parshwa Nandu (BS, Dwarkadas J Sanghvi College of Engineering)

## TALKS

Automated Assertion Generation and Regression Testing for Machine Learning Notebooks *ASE 2026*

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms *DSN 2023*

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms. *CCS 2023*

## OPEN-SOURCE CONTRIBUTIONS

**SwarmFuzz**: https://github.com/DependableSystemsLab/SwarmFuzz

**NBTest**: https://github.com/seal-research/NBTest